Busting the Myth: PIRLS' SLR Scale Isn't Valid for Country-to-Country Comparison

Within and Beyond Borders? Intrinsic Reading Motivation in PIRLS

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Introduction

Intrinsic reading motivation is believed to be a powerful force that drives children to engage in out-of-school reading, enhancing their reading competencies [1,2]. Among instruments designed to measure this motivation, the PIRLS' Students Like Reading scale (P-SLR) stands out as it is used as a tool for international comparisons of Year 4 students every five years. However, the scale's psychometric validity remains underexamined in prior work [3–5], particularly regarding its invariance across countries and student subgroups.

This study seeks to rigorously examine the psychometric properties of the P-SLR scale in two countries: Denmark and Norway. The aims are to

1. aim: Evaluate if the P-SLR scale is measurement invariant across similar countries (main)

2. aim: Evaluate the construct validity within countries to identify potential deviations (secondary)

Table 1. Items in the P-SLR scale.

#	Wording	Scale
1	I read for fun	Never - Everyday (1-4)
2	I read to find out things I want to learn	Never - Everyday (1-4)
3	I like talking about what I read with other people	Disagree - Agree (1-4)
4	I would be happy if someone gave me a book as a present	Disagree - Agree (1-4)
5	I think reading is boring	Disagree - Agree (1-4)
6	I would like to have more time for reading	Disagree - Agree (1-4)
7	I enjoy reading	Disagree - Agree (1-4)
8	I learn a lot from reading	Disagree - Agree (1-4)
9	I like to read things that make me think	Disagree - Agree (1-4)
10	I like it when a book helps me imagine other worlds	Disagree - Agree (1-4)

Methods

Data Source: PIRLS 2016 cycle.

Sample: Denmark (n=3508), Norway (n=4232). Given the shared socio-economic, schooling, and linguistic attributes of these countries, they are ideal for testing invariance across borders. The pooled sample consists of 2000 randomly sampled individuals from each country (n=4000).

Model: Rasch Partial Credit Model [6].

Item analysis: All analyses done with the `eRm` and `iarm` package [7,8].

Table 2. Item analysis.

Type	Test	Method used
Global Tests	Homogeneity	Andersen's CLR test
	Invariance	Andersen's CLR test
Item Tests	Local Dependence	Partial correlation: Goodman-Kruskal γ
	DIF	Partial correlation: Goodman-Kruskal γ
	Item Fit	Observed and expected item-rest correlation γ

Note:
Exogeneous variables for DIF: Gender, test language and parental education

Results

Table 3. Global tests-of-fit and Invariance for the P-SLR scale.

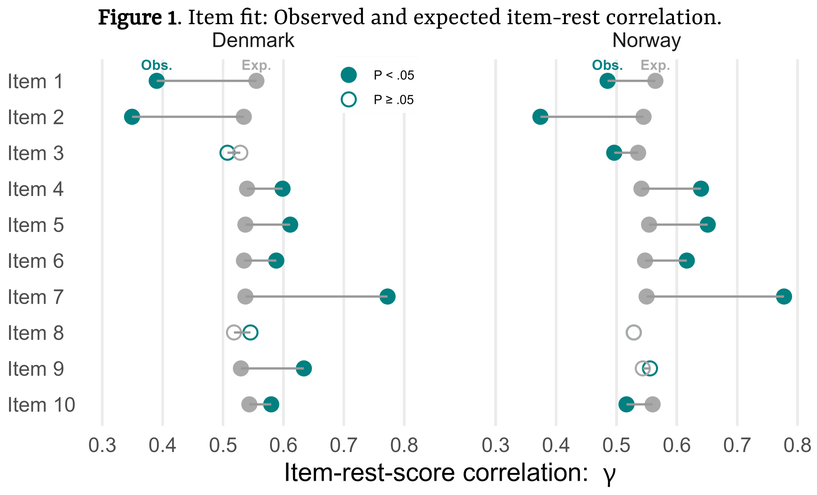
	Pooled			Denmark			Norway		
	CLR	df	p	CLR	df	p	CLR	df	p
Homogeneity[a]	1020	29	<.001	715	29	<.001	855	29	<.001
Invariance[b]									
Country	397	29	<.001						
Gender	179	29	<.001	129	29	<.001	235	29	<.001
Test language at home	105	29	<.001	110	29	<.001	76	29	<.001
Parental education	190	29	<.001	110	29	<.001	187	29	<.001

Note:
CLR: conditional likelihood-ratio test, df: degrees of freedom, p: p-value.
[a] Tests the hypothesis that all item parameters are consistent across low or high scorers.
[b] Tests the hypothesis that all item parameters are consistent across subgroups.

Key findings from the global tests are:
- Pooled data: Model fit rejected, non-equal parameters
- Country Invariance: None, parameters vary by country
- Within-country: Model fit rejected, issues beyond country

Lack of country invariance and model misfit could stem from many sources, e.g. item misfit (in different countries), DIF or multidimensionality...

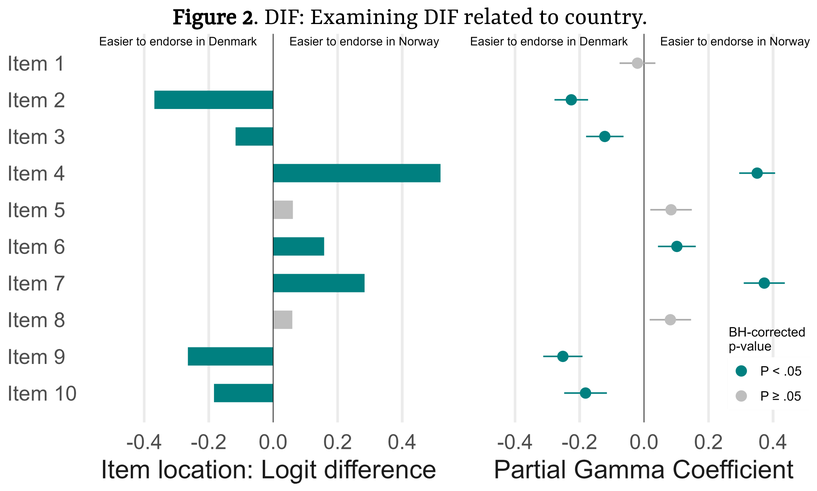
Figure 1. Item fit: Observed and expected item-rest correlation.



Note: The fit of individual items was tested by comparing the observed and the expected item-rest-score correlations under the Rasch model [9]. P-values for the comparison are adjusted for multiple testing with the Benjamini-Hochberg method [10]. Conditional In- and Outfit statistics also show widespread issues with item misfit (see the appendix).

Analysis of item fit shows...
- Widespread issues in each country (8 items misfit)
- Item misfit alike across countries; may not be main cause
- Results confirm within-country issues (see also Table 3)

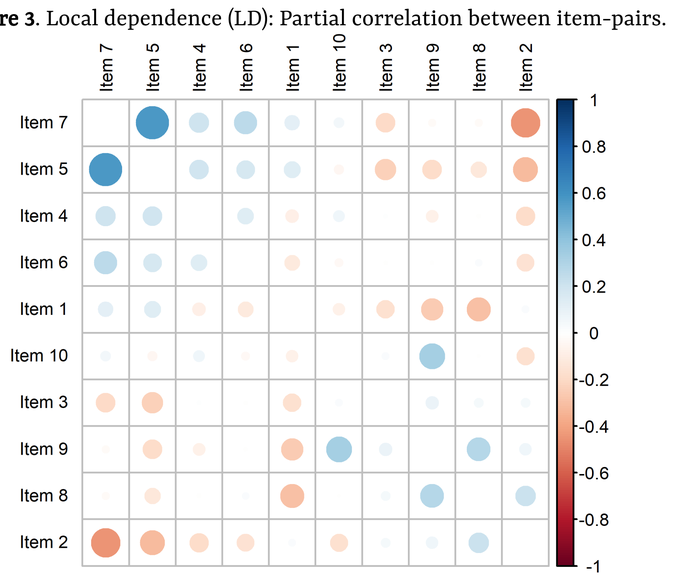
Figure 2. DIF: Examining DIF related to country.



Note: The left panel displays differences in item locations between countries, calculated using item 1 as an anchor and splitting the remaining items by country. The right panel shows partial Goodman-Kruskal gamma coefficients that tests for a significant association between item score and country, controlling for the scale score, with 95% confidence intervals as horizontal lines. The p-value is corrected for multiple comparisons using the Benjamini and Hochberg method [10].

Analysis of cross-country DIF reveals...
- 7 out of 10 items show significant DIF, direction varies
- Largest difference ~0.5 logits (location range: -0.4 to 1).
- Overall invariance not simply due to measurement error

Figure 3. Local dependence (LD): Partial correlation between item-pairs.



Note: The figure displays whether there is any correlation left between two items once the underlying factor has been taken into account. Partial Gamma coefficients between pairs of items controlled for the rest score is used to assess this requirement. The figure reports the mean partial gamma coefficient, as the calculation of the rest score for each item-pair involves excluding either the first or the second item. Items are ordered from a dimensional angle in line with Schiefele et al.'s idea [2], which differentiates between two separate intrinsic reading motivation constructs: activity and object-specific motivation.

Analysis of LD shows...
- Multiple instances of positive and negative LD
- Positive LD within item clusters, negative LD between
- Sign of subscales? Activity and object-specific motivation

Conclusion

There is little evidence that the P-SLR scale is invariant across country. The results show:
- **No country invariance**: Even among similar countries
- **In-country misfit**: Suggests issues unrelated to country
- **Multiple dimensions?** LD hints at multidimensionality

Future directions
- Does multidimensionality sort lack of invariance?
- Reassess quality of misfitting items?
- Generate new items to instrument

Resources

All underlying results are available here: Link

Or scan the QR code in the bottom

The poster has been made with the `posterdown` package [11]

References

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